

DAY 4

NUMPY

DATA VISUALISAION

Data visualisation is a technique which can perform the data to visualise for getting the graphs, plots and charts.

```
!pip install numpy
```

```
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
```

```
import numpy as np
```

```
print(np.__version__)
```

```
2.0.2
```

```
a=np.ndarray([1,2,3,4])
```

```
print(a)
```

```
[[[-0.25 -0.5   0.   -0.25]
  [ 0.25 -0.5   0.5  -0.25]
  [ 0.25  0.    0.5   0.25]]

 [[ 0.25  0.5   0.    0.25]
 [-0.25  0.5  -0.5   0.25]
 [-0.25  0.   -0.5  -0.25]]]
```

```
A=np.array([1,2,3,4,5])
```

```
print(A)
```

```
[1 2 3 4 5]
```

```
b=np.array([[1,2,3],[4,5,6],[4,6,7]])
```

```
print(b)
```

```
[[1 2 3]
 [4 5 6]
 [4 6 7]]
```

```
c=np.array([
    [
        [1,2],
        [3,4]
    ],
    [
        [5,6],
        [7,8]
    ]
])
```

```

    ]]
)
print(c)

[[[1 2]
   [3 4]]

 [[5 6]
   [7 8]]]

a=np.zeros([3,3])
print(a)

[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]

a=np.ones([3,3])
print(a)

[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]

a=np.full([3,3],9)
print(a)

[[9 9 9]
 [9 9 9]
 [9 9 9]]

def greet():
    """
    This is vishnu
    """
    print("Hello")
print(greet())
print(greet.__doc__)

Hello
None

    This is vishnu

a=np.full([6,6],9)
print(a)

[[9 9 9 9 9 9]
 [9 9 9 9 9 9]
 [9 9 9 9 9 9]
 [9 9 9 9 9 9]
 [9 9 9 9 9 9]
 [9 9 9 9 9 9]]

```

```

b=np.ones([3,3])
print(b)

[[1.  1.  1.]
 [1.  1.  1.]
 [1.  1.  1.]]

c=np.zeros([3,3])
print(c)

[[0.  0.  0.]
 [0.  0.  0.]
 [0.  0.  0.]]

d=np.eye(3,3)
print(d)

[[1.  0.  0.]
 [0.  1.  0.]
 [0.  0.  1.]]

m=np.arange(2,10,3)
print(m)
n=np.arange(3,12,8)
print(n)

[2  5  8]
[ 3 11]

np.linspace(2,4,8)

array([2.          , 2.28571429, 2.57142857, 2.85714286, 3.14285714,
       3.42857143, 3.71428571, 4.          ])

#range from (0,1)
np.random.rand(3,3)

array([[0.14998335, 0.22453972, 0.44176543],
       [0.24435698, 0.94865296, 0.73754322],
       [0.61161441, 0.83589698, 0.32255628]])

#normal distribution
np.random.randn(3,3)

array([[ 0.21965052, -0.07910779,  1.77011274],
       [-0.83526075,  0.80621675, -0.60234833],
       [ 1.0237581 ,  0.41088594, -1.74200798]])

#random integers
np.random.randint(1,10,size=(3,3))

array([[7, 8, 5],
       [5, 8, 2],
       [5, 5, 2]])

```

```
#uninitialize (garbage value)
```

```
np.empty((3,3))
```

```
array([[ 0.21965052, -0.07910779,  1.77011274],  
       [-0.83526075,  0.80621675, -0.60234833],  
       [ 1.0237581 ,  0.41088594, -1.74200798]])
```

```
arr=np.array([[1,2,3,4,5],[2,3,4,5,7],[0,9,8,78,6],[1,2,3,4,5],[9,7,6,5,4]])
```

```
print(arr.shape)
```

```
print(arr.ndim)
```

```
print(arr.dtype)
```

```
print(arr.size)
```

```
print(arr.itemsize)
```

```
print(arr.nbytes)
```

```
(5, 5)
```

```
2
```

```
int64
```

```
25
```

```
8
```

```
200
```

```
#indexing
```

```
a=np.array([10,20,30,40,50])
```

```
print(a[0])
```

```
print(a[1])
```

```
print(a[-2])
```

```
print(a[-1])
```

```
print(a[-4])
```

```
print(a[-5])
```

```
10
```

```
20
```

```
40
```

```
50
```

```
20
```

```
10
```

```
#slicing
```

```
print(a[1:4])
```

```
print(a[:3])
```

```
print(a[::2])
```

```
print(a[::-1])
```

```
print(a[:-1])
```

```
[20 30 40]
```

```
[10 20 30]
```

```
[10 30 50]
```

```
[50 40 30 20 10]
```

```
[10 20 30 40]
```

```
#2D indexing
```

```
arr = np.array([[1, 2, 3, 4, 5], [2, 3, 4, 5, 7],[2,1,3,4,5]])
```

```
print(arr[0,1])
```

```
print(arr[0,3])
print(arr[0,:]) #first row
print(arr[:,1]) #second column
print(arr[0:1, 1:2])
```

```
2
4
[1 2 3 4 5]
[2 3 1]
[[2]]
```

```
a= np.array([10,24,56,10,19])
#boolean mask
mask = a>15
print(mask)
print(a[mask])
```

```
#short hand
print(a[a>15])
```

```
#fancy indexing
print(a[[2,3]])
```

```
[False  True  True False  True]
[24 56 19]
[24 56 19]
[56 10]
```